CCSA104

Security Policy Management

Practical Assignment: Lab 4

Managing Check Point Security Policies

Access Control rulebase, sections, Hide NAT and the rule-order experiment

Course / module	CCSA104
Assignment	Practical Assignment: Lab 4 (Parts A and B)
Student name	Akinsola, Gideon Oluwaferanmi
Student ID	C11/26/CCSA/17099
Submission date	8th November, 2026
Lab environment	Check Point SmartConsole R82; A-SMS; policy package Standard; target A-GW-Cluster

Table of Contents

1. Introduction 3
1.1 Purpose of the practical 3
1.2 Objectives of managing Check Point Security Policies 3
1.3 The laboratory environment 3
1.4 Status key used throughout this report 5
2. Task 1: Exploration of the Default Policy 7
2.1 The Standard policy package 7
2.2 The default Cleanup rule 7
2.3 Rulebase columns: Hits, Track and Install On 8
2.4 Available policy packages 9
2.5 Global Properties 10
3. Task 2: Building the Access Control Rulebase 12
3.1 Object preparation 12
3.2 The completed rulebase 13
3.3 Purpose of each rule 14
3.4 Rule order and the first-match principle 15
3.5 Proposed configuration: Rule 4, Internal to DMZ 15
4. Task 3: Organising the Rulebase into Sections 17
4.1 The four sections 17
4.2 Purpose of rulebase organisation 18
4.3 Expand and collapse demonstration 18
5. Task 4: Configuring Hide NAT 20
5.1 The A-INT-NET NAT configuration 20
5.2 The automatically generated NAT rules 21
5.3 Significance: automatic versus manual rules, many-to-one translation and ports 22
5.4 Configuration evidence versus live verification 23
6. Task 5: Rule Order Experiment and Installation History 24
6.1 The temporary rule 24
6.2 Stages performed 24
6.3 Deletion and restoration of the original rulebase 28
6.4 The installation failure 28
6.5 Implications of the expired certificates 30
6.6 Three distinct stages: publish, install, verify 31
6.7 The rule-order experiment in principle 31
7. Challenges, Limitations and Proposed Configurations 33
7.1 Gaps and inconsistencies in the supplied brief 33
7.2 Proposed configurations 34
7.3 Use of the Standard policy package 34
7.4 Expired certificates and the unsuccessful installation 34
7.5 What could not be verified 34

8. Conclusion 35
9. Part B: Theory Questions 36
Section B1: Multiple choice 36
Section B2: Short answers 36
Q6. Explicit rules, Implied rules and the Implicit Cleanup Action 36
Q7. Why rule order matters 37
Q8. Saving versus publishing, and installing 37
Q9. Global Properties compared with Policy Packages 38
Q10. Security Zones 38
Section B3: Scenario questions 39
Q11. An Accept Any rule inserted above the Stealth rule 39
Q12. Staff cannot reach the Internet after Hide NAT is configured 39
10. References 41

1. Introduction

1.1 Purpose of the practical

This report documents the practical work I carried out for the CCSA104 Practical Assignment (Lab 4, Security Policy Management). The assignment asks me, acting as security administrator for the fictional company Orion Logistics, to build an Access Control rulebase that follows the Positive Control Model, organise it with sections, configure Hide NAT, and then publish, install and verify the policy. In it I record what I actually configured in SmartConsole, what I observed, and where I could not complete or verify the work.

The report is deliberately explicit about the difference between configuration and enforcement. I created and published all of the configuration described below on the Security Management Server (A-SMS). Every attempt I made to install the policy on the gateway cluster (A-GW-Cluster) failed, so I do not claim that the gateway enforced any rule or that I tested any network traffic.

1.2 Objectives of managing Check Point Security Policies

The assignment brief sets four objectives, which I address in turn:

- **Explain the purpose of a Security Policy and apply the Positive Control Model.** A Check Point Access Control policy is created in SmartConsole, stored on the Security Management Server and enforced by the Security Gateways. Under the Positive Control (whitelist) Model only explicitly permitted traffic is allowed and everything else is dropped and logged.
- **Configure the essential policy elements.** These are rules, the Stealth rule, the Cleanup rule, sections and NAT (Tasks 2, 3 and 4). The brief's overview also mentions security zones, but no practical task in the supplied brief configures them (see Section 7).
- **Follow the workflow of create, publish, install and verify.** I completed the first two stages. The install stage failed and I could not reach the verify stage.
- **Use features that support policy management.** These are Global Properties, policy packages and Installation History (Tasks 1 and 5).

1.3 The laboratory environment

I carried out the work in Check Point SmartConsole (R82 gateways, as shown in the Version column of Figure 1) connected to the Security Management Server **A-SMS** (10.1.1.101). The intended enforcement point was the gateway cluster **A-GW-Cluster** (10.1.1.1), whose members are **A-GW-01** (10.1.1.2) and **A-GW-02** (10.1.1.3). I used the policy package **Standard** for the practical; it initially contained only the default Cleanup rule. A second package, **Alpha-Standard**, also exists in the environment and is the Access Control policy shown against A-GW-Cluster (Figure 1).

The brief gives example object names and addresses and instructs me to use the addressing of my own lab. The mapping I adopted is shown in Table 1.

Table 1: Mapping of the brief's object names to the objects used in the lab

Brief reference	Object used in SmartConsole	Value	Where the value is evidenced
Internal_LAN	A-INT-NET	192.168.11.0/24	My lab record; object name visible in Figures 7 and 11
DMZ_Net	A-DMZ-NET	192.168.12.0/24	My lab record; not used directly in any rule
Admin_PC	A-GUI	10.1.1.201	My lab record; object name visible in Figure 7
Web_Server	Student-Web-Server	192.168.12.150	My lab record; object name visible in Figure 7
Mail_Server	Mail_Server (new, proposed)	192.168.12.151	Figure 6 (New Host dialog)
External_DNS	Azure-External DNS	168.63.129.16	My lab record; object name visible in Figure 7
Gateway	A-GW-Cluster	10.1.1.1	Figure 1

Where the table says "my lab record", the address comes from the lab topology and is not legible in the screenshots, so I report it as recorded rather than independently confirmed.

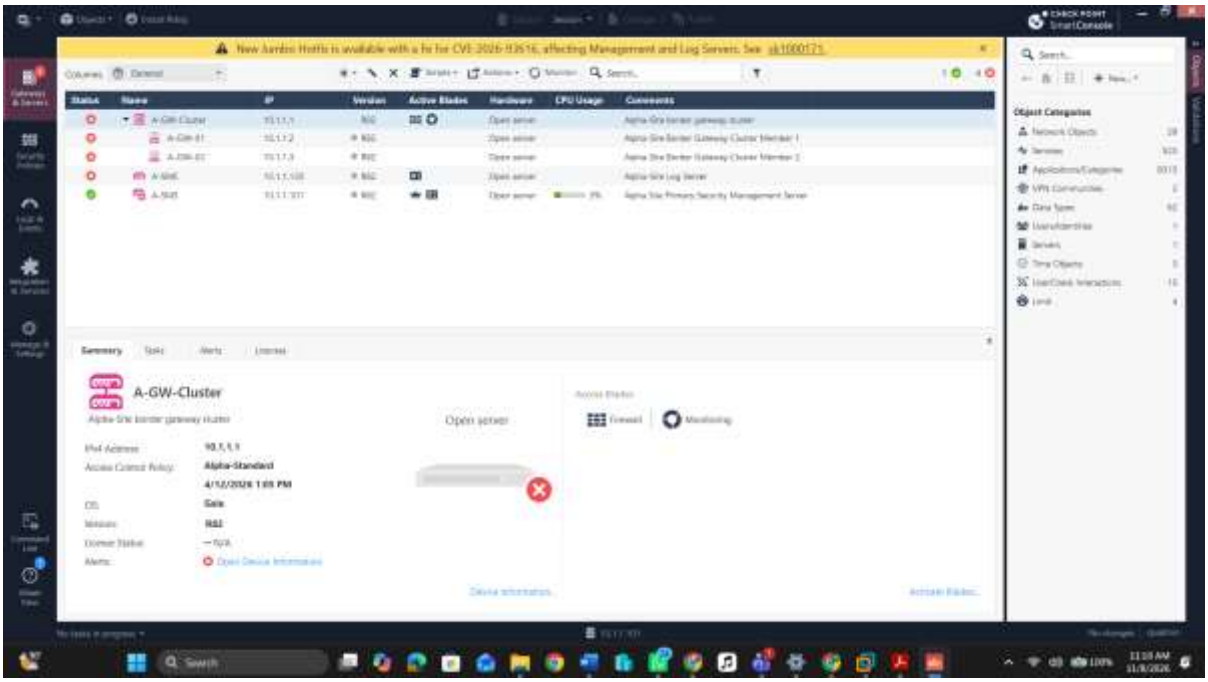


Figure 1: *SmartConsole Gateways & Servers view. A-GW-Cluster (10.1.1.1), A-GW-01 (10.1.1.2), A-GW-02 (10.1.1.3) and A-SME (10.1.1.130) show a red status indicator; A-SMS (10.1.1.101) shows a green one. The summary pane shows Alpha-Standard as the Access Control Policy of A-GW-Cluster. The screenshot does not state the cause of the red indicators.*

1.4 Status key used throughout this report

To avoid overstating my work, I classified each activity as one of four statuses. Table 2 summarises the outcome of the practical on that basis; the detail follows in Sections 2 to 7.

Table 2: Summary of outcomes

Activity	Status	Basis
Exploration of the Standard policy package (Task 1)	Completed	Figures 2 to 5
Mail_Server host object	Completed as a **proposed** object	Figure 6. No mail server was deployed or tested at 192.168.12.151.
Eight-rule Access Control rulebase	Completed (configuration only)	Figure 7, published on A-SMS
Rule 4, Internal to DMZ	**Proposed** configuration	The brief does not define it (Section 3.5)

Activity	Status	Basis
Four rulebase sections; expand and collapse	Completed	Figures 8 to 10
Hide NAT on A-INT-NET and automatic NAT rules	Completed (configuration only)	Figures 11 and 12. Translation not observed.
TEMP Block Internal: create, move, delete, restore order	Completed (configuration only)	Figures 13 to 18
Policy installation on A-GW-Cluster (two attempts)	**Attempted, unsuccessful**	Figures 14, 17, 19, 20
Traffic tests before and after the rule move	**Not performed**	Not possible without a successful installation
Gateway enforcement, logging of live traffic, NAT translation	**Not verified**	Environmental limitation: expired gateway VPN certificates

2. Task 1: Exploration of the Default Policy

2.1 The Standard policy package

After logging in to SmartConsole and opening the Security Policies view, I recorded the policy package in use as **Standard** (the open tab in Figures 2 and 3). The Access Control section of the left-hand navigation shows Policy and NAT; Threat Prevention and HTTPS Inspection sections are also present. The Standard package contained a single rule, the default Cleanup rule, which gave me the starting point for my practical work.

2.2 The default Cleanup rule

Table 3 records the values of the default Cleanup rule as displayed in Figures 2 and 3, before I made any change.

Table 3: Default Cleanup rule values (Standard package, before changes)

Field	Value
No.	1
Name	Cleanup rule
Source	Any
Destination	Any
VPN	Any
Services & Applications	Any
Action	Accept
Track	Log
Install On	Policy Targets
Created by / date created	System, 10/15/2024 12:14 PM
Hit count	405 (100%, Very high)

Two points about this rule are worth stating. First, its Action is **Accept**. A rule that matches Any source, Any destination and Any service with Action Accept is the opposite of the Positive Control Model: anything that reaches it is permitted. The assignment therefore requires it to be turned into a **Drop** rule so that unmatched traffic is blocked. Second, its Track value was already **Log** in the default package (Figure 2), so the brief's instruction to set Track to Log required no change to that field. The changes I actually made are in Section 3.4.

The hit count of 405 shows that the rule had already matched traffic in the environment before the exercise, which is consistent with a catch-all rule at the end of an otherwise empty rulebase. It should not be read as evidence about the current gateway state.

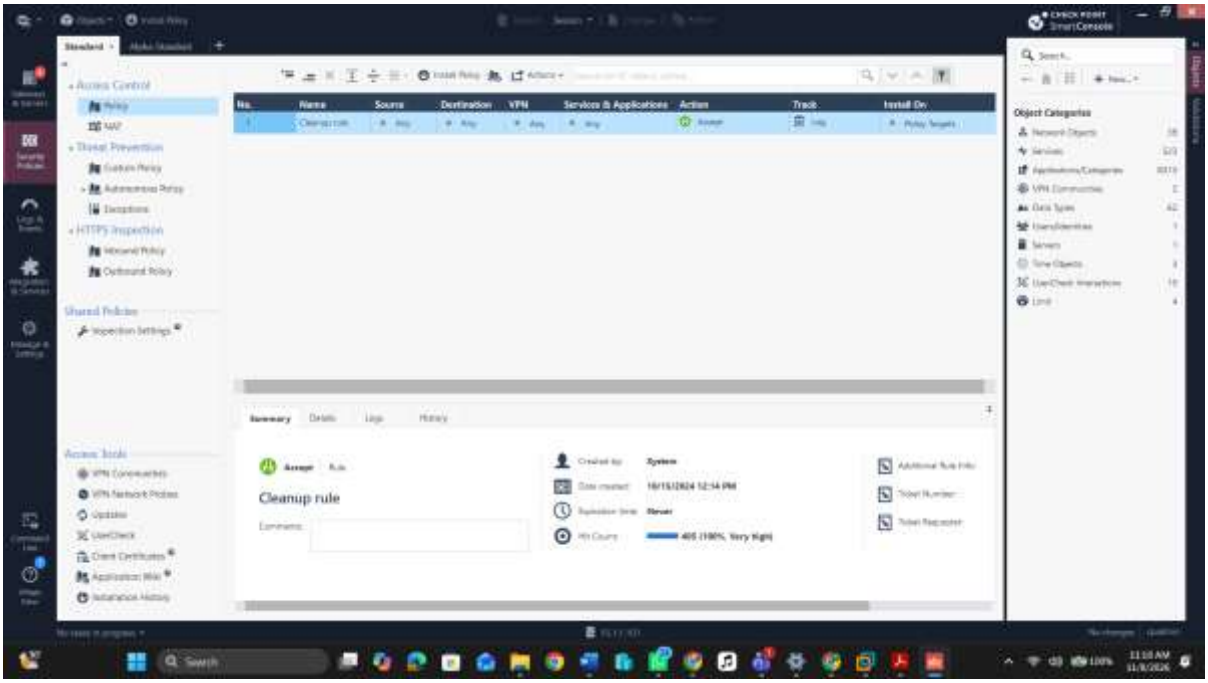


Figure 2: *The default Standard policy package: a single rule, Cleanup rule, with Source, Destination and Services Any, Action Accept, Track Log and Install On set to Policy Targets. The Summary pane shows Created by System and a hit count of 405.*

2.3 Rulebase columns: Hits, Track and Install On

The brief asks for the Hits, Track and Install On columns to be shown, one column hidden and then shown again. Figure 2 already displays Track and Install On, and Figure 3 additionally displays the **Hits** column, which shows the rule's hit count (405) as a bar. The **Track** column controls what is recorded when a rule matches (here, Log), and the **Install On** column controls which gateways enforce the rule; the default value, Policy Targets, means the gateways that are the targets of the policy package. I did not capture the hide-and-restore step itself; my screenshots show only the end state, so I do not claim it as evidenced.

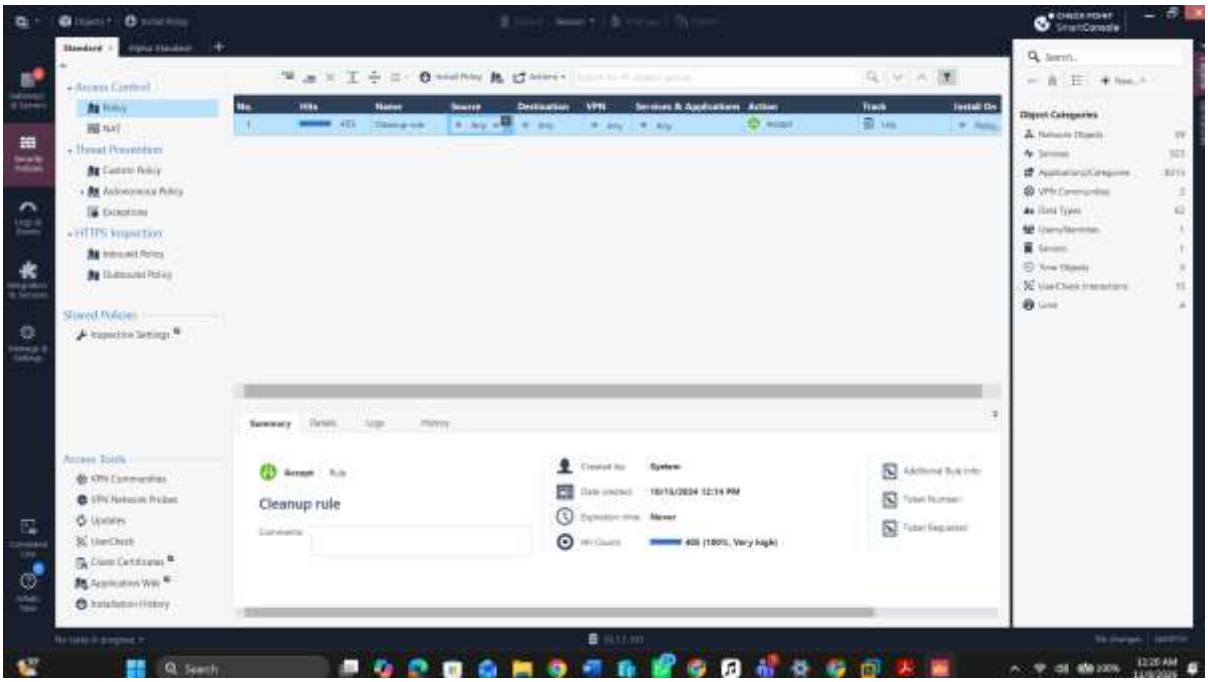


Figure 3: *The Standard rulebase with the Hits column displayed (405) alongside the Track and Install On columns.*

2.4 Available policy packages

Opening **Manage policies and layers** lists two policy packages (Figure 4):

- **Alpha-Standard**: Access Control and Threat Prevention enabled; Policy Targets A-GW-Cluster.
- **Standard**: Access Control and Threat Prevention enabled; Policy Targets **All gateways**.

This is significant for the later installation failure. The cluster's current Access Control policy is shown as Alpha-Standard (Figure 1), whereas I built my practical work in Standard. The Standard package targets all gateways, so I gave each rule in it an explicit Install On value of A-GW-Cluster (Section 3.4).

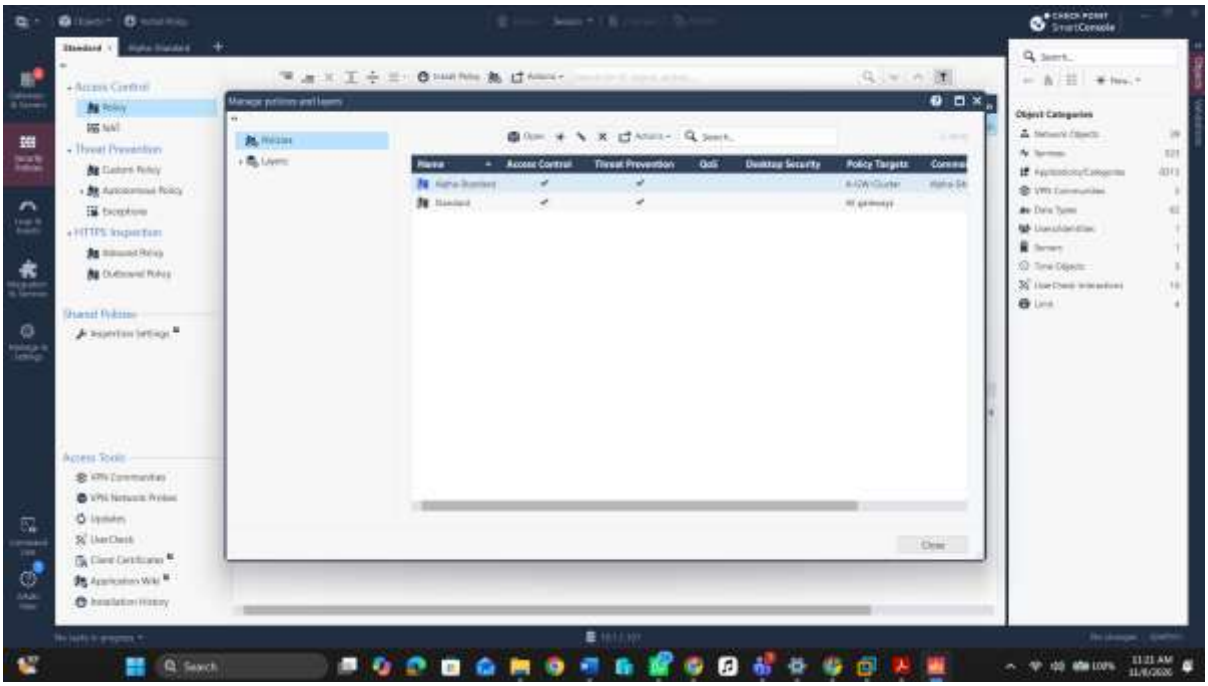


Figure 4: *Manage policies and layers: the two policy packages, Alpha-Standard (target A-GW-Cluster) and Standard (target All gateways).*

2.5 Global Properties

Global Properties holds settings that apply across the whole management domain rather than to a single rule or package. The category tree on the left of the Global Properties window in Figure 5 includes **FireWall**, **NAT** (the label is truncated in the screenshot), **Authentication**, **VPN**, **Identity Awareness**, **Remote Access**, **User Directory**, **QoS**, **Log and Alert** and **Advanced**, among others. The page opened in Figure 5 is the FireWall page, which controls the **implied rules**: predefined rules that permit internal Check Point communication, each with a configurable position (for example First or Before Last). The **Log Implied Rules** option is unticked in the screenshot, so implied rule matches are not logged by default. I did not change any setting in Global Properties.

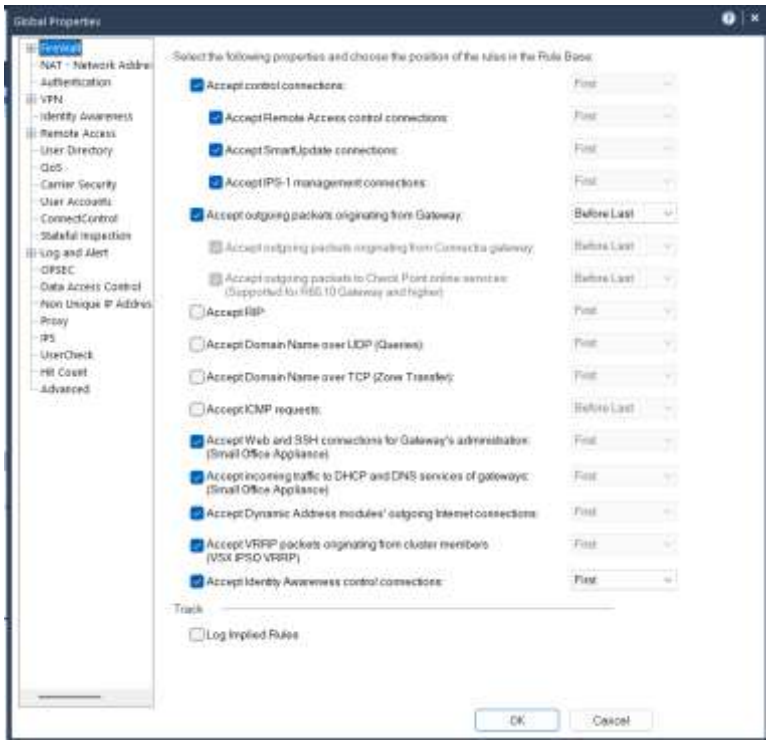


Figure 5: *Global Properties, FireWall page: implied-rule options with their rulebase positions, and the Log Implied Rules option (unticked).*

3. Task 2: Building the Access Control Rulebase

3.1 Object preparation

Most of the required objects already existed in the environment under lab-specific names (Table 1). The exception was the mail server: no Mail_Server object was available. I therefore created a new host object with the name required by the brief.

Table 4: The proposed Mail_Server host object

Property	Value (Figure 6)
Name	Mail_Server
Object type	Host
IPv4 address	192.168.12.151
Comment	Proposed Mail Server in Alpha-Standard DMZ

I chose the address from the DMZ subnet (192.168.12.0/24), adjacent to the existing web server at 192.168.12.150. **This is a proposed policy object only.** It exists so that the rules in the brief can reference a mail server by name. I did not deploy or verify a mail server at 192.168.12.151, and nothing in this report should be read as showing that the address responds or hosts an SMTP service.



Figure 6: *New Host dialog for Mail_Server (192.168.12.151) with the comment "Proposed Mail Server in Alpha-Standard DMZ", shown over the Object Explorer.*

3.2 The completed rulebase

I arranged eight rules in the Standard package so that the final order matches the brief's table. I inserted seven rules above the existing Cleanup rule and kept the Cleanup rule at the bottom. Every rule has **A-GW-Cluster** in the Install On column. Table 5 and Figure 7 show the configuration as built; I captured Figure 7 before adding the sections of Task 3.

Table 5: Access Control rulebase in the Standard policy package (as configured)

No.	Name	Source	Destination	Services	Action	Track	Install On
1	Admin Access	A-GUI	A-GW-Cluster	ssh, https	Accept	Log	A-GW-Cluster
2	Stealth Rule	Any	A-GW-Cluster	Any	Drop	Log	A-GW-Cluster
3	Inbound to DMZ	Any	Student-Web-Server Mail_Server	http, https, smtp	Accept	Log	A-GW-Cluster
4	Internal to DMZ (**proposed**)	A-INT-NET	Student-Web-Server Mail_Server	http, https, smtp	Accept	Log	A-GW-Cluster
5	DNS Out	A-INT-NET	Azure-External DNS	dns	Accept	Log	A-GW-Cluster
6	Internal Web Out	A-INT-NET	Any	http, https	Accept	Log	A-GW-Cluster
7	SMTP Out	Mail_Server	Any	smtp	Accept	Log	A-GW-Cluster
8	Cleanup rule	Any	Any	Any	Drop	Log	A-GW-Cluster

Note on the DNS service: the Services column in Figure 7 shows a single service object named **dns**. The brief specifies DNS over both UDP and TCP; the screenshot does not show whether the object covers both protocols, so I record the object as displayed and do not assert its protocol membership.

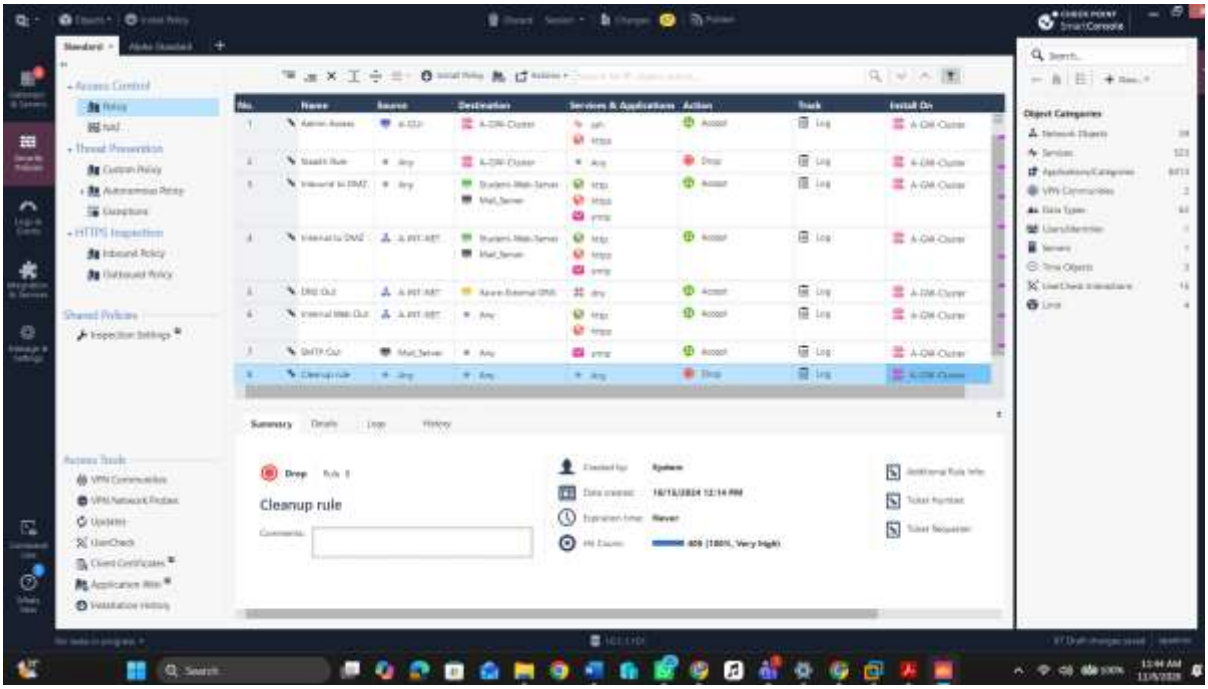


Figure 7: *The eight-rule Access Control rulebase in the Standard package before sections were added (67 draft changes saved). Rule 8, the Cleanup rule, is selected and shows Drop, Log and Install On A-GW-Cluster.*

3.3 Purpose of each rule

Rule 1, Admin Access. Permits the administrator workstation A-GUI to reach the gateway cluster over SSH and HTTPS, so that the gateway can still be administered (command line and Gaia web interface) after the Stealth rule is in force. It is deliberately narrow: one source, one destination and two services. Communication between the management server and the gateways for policy installation is handled by Check Point's implied control-connection rules (Figure 5), not by this rule.

Rule 2, Stealth Rule. Drops and logs any connection addressed to the gateway cluster that was not already permitted by Rule 1. It stops hosts probing the firewall itself and records the attempt. It must sit directly after the management rule and above the general rules (Section 3.4).

Rule 3, Inbound to DMZ. Allows any source to reach the web server and the (proposed) mail server over HTTP, HTTPS and SMTP. This publishes the DMZ services, and nothing else in the DMZ, to external users.

Rule 4, Internal to DMZ. Proposed rule giving the internal network A-INT-NET access to the same DMZ servers and services (Section 3.5).

Rule 5, DNS Out. Allows internal hosts to query only the named external DNS server, Azure-External DNS, rather than any resolver on the Internet. This limits both exposure and data-exfiltration paths.

Rule 6, Internal Web Out. Allows internal users to browse the Internet over HTTP and HTTPS.

Rule 7, SMTP Out. Allows the mail server, and only the mail server, to send mail to any destination over SMTP.

Rule 8, Cleanup rule. Drops and logs everything not matched above, so that the policy implements the Positive Control Model and unmatched attempts are visible in the logs.

3.4 Rule order and the first-match principle

The gateway evaluates the Access Control rulebase from the top and applies the **first** rule that matches the connection; later rules are not examined for that connection. Order is therefore part of the policy's meaning, not a presentation choice. Three features of this rulebase depend on it:

- **Admin Access above Stealth Rule.** The Stealth rule matches any source addressed to the cluster. If it came first, it would also match the administrator's SSH and HTTPS sessions and drop them, locking the administrator out. Placing the specific management rule above the general Stealth rule avoids this. This arrangement is the same as the basic policy shown in Check Point's Access Control guidance, where the Stealth rule follows the administrator-access rule (Check Point Software Technologies, n.d.a).
- **Specific permissions above the Cleanup rule.** The Cleanup rule matches everything, so it must be last. Any rule placed beneath it could never be reached.
- **Specific rules before general ones.** The experiment in Task 5 (Section 6) examines this principle directly.

I modified the Cleanup rule as follows. It had Action Accept and Install On set to Policy Targets in the default package (Figure 2). I changed it to **Action Drop**, **Track Log** (already Log) and **Install On A-GW-Cluster**, and left it at the bottom of the rulebase (Figure 7). Check Point's guidance likewise recommends an explicit Cleanup rule as the last rule of each layer (Check Point Software Technologies, n.d.a). Without an explicit rule, unmatched traffic is handled by the layer's implicit cleanup rule, whose default action is Drop but which is not shown in the rulebase (Check Point Software Technologies, n.d.b). SmartConsole itself warns in Figure 10 that unmatched traffic "will be dropped and not logged" when an explicit logging cleanup rule is missing, which is why the explicit rule with Track set to Log is used.

3.5 Proposed configuration: Rule 4, Internal to DMZ

The brief's rulebase table gives Rule 4 the name "Internal to DMZ" and states that it is "added in Task 5". However, Task 5 in the supplied brief is the rule-order experiment and does not describe any such rule, and no other part of the brief gives its Source, Destination, Services, Action or Track. Its definition was therefore **not specified** in the brief. I adopted the following configuration as an **assumed, proposed configuration**:

Field	Proposed value
Name	Internal to DMZ

Field	Proposed value
Source	A-INT-NET
Destination	Student-Web-Server, Mail_Server
Services	http, https, smtp
Action / Track / Install On	Accept / Log / A-GW-Cluster

Rationale. The rule's name implies traffic from the Internal network to the DMZ. The services mirror Rule 3, because those are the services the DMZ servers exist to provide. Staff need to browse the intranet web server and send mail through the mail server, which gives the rule a plausible purpose. I placed it immediately after Rule 3, inside the DMZ Services section, as the brief's final-position table requires.

Critical observation. As configured, Rule 4 is redundant. Rule 3 has Source Any, so it already matches traffic from A-INT-NET to the same destinations and services, and with the same Action and Track. Because the gateway applies the first matching rule, internal traffic to the DMZ servers would be accepted and logged under Rule 3 and would never reach Rule 4. Rule 4 would only become effective if Rule 3 were narrowed (for example to an external-only source) or if Rule 4 were given different properties. Because no installation succeeded, I did not observe this behaviour in logs. I make this point rather than conceal it, since it follows directly from the first-match principle and from the brief's own definition of Rule 3.

4. Task 3: Organising the Rulebase into Sections

4.1 The four sections

I added the section titles with **New Section Title** and placed each rule under the appropriate title, as the brief specifies. The result is shown in Figure 8, where SmartConsole displays the rule range of each section in brackets.

Table 6: Rulebase sections

Section title	Rules (Figure 8)	Purpose
Management and Protection	1 to 2 (Admin Access, Stealth Rule)	Protects and administers the gateway itself
DMZ Services	3 to 4 (Inbound to DMZ, Internal to DMZ)	Access to the DMZ web and mail servers
Outbound Access	5 to 7 (DNS Out, Internal Web Out, SMTP Out)	Connections leaving the internal network or DMZ
Cleanup	8 (Cleanup rule)	Default drop and log

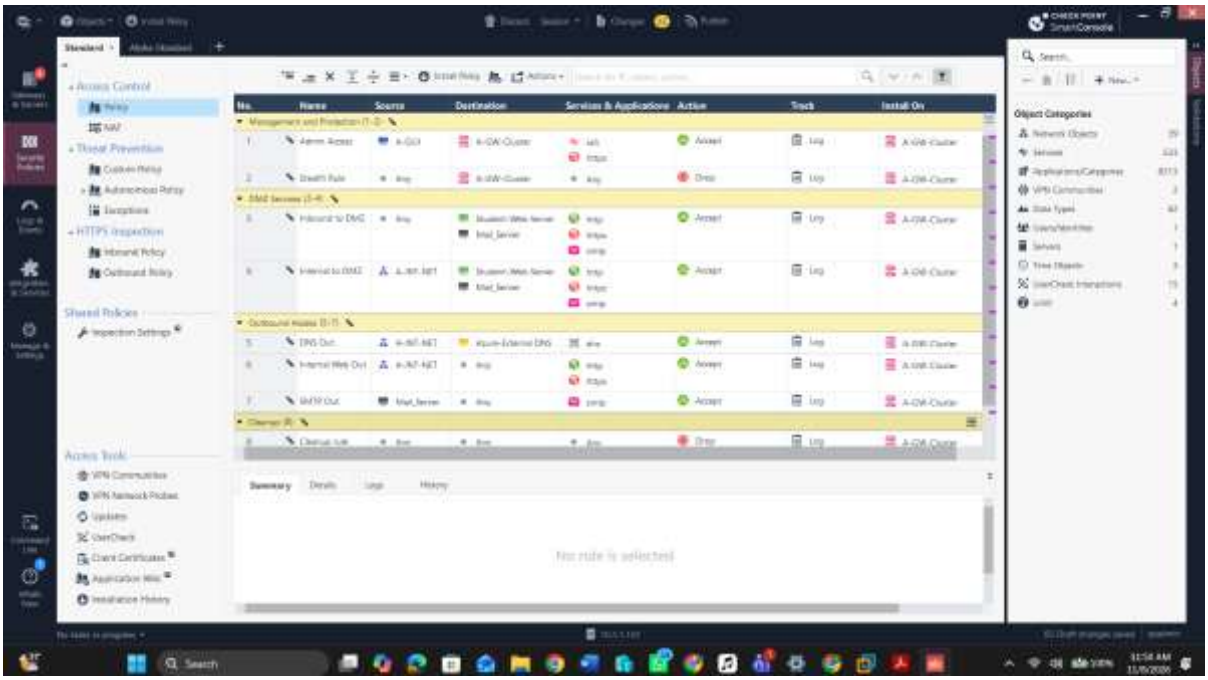


Figure 8: *The rulebase with all four sections expanded: Management and Protection (1-2), DMZ Services (3-4), Outbound Access (5-7) and Cleanup (8). 82 draft changes are saved at this point.*

4.2 Purpose of rulebase organisation

A rulebase of eight rules is easy to read without sections, but production rulebases frequently contain hundreds. Sections group rules by function so that an administrator can find the management rules, the DMZ rules or the outbound rules without scanning the whole list, and so that review and change control can be discussed section by section. The titles here follow the logical flow of the policy: protect the gateway, publish the DMZ, allow outbound access, then drop the remainder.

Sections are an organisational device only. They do not independently change any access-control decision. The gateway still evaluates the rules from top to bottom and applies the first match, exactly as it would with no sections at all, and a section title is not itself a rule that matches or acts on traffic. Moving a rule from one section to another changes where it sits in the order (and therefore may change the outcome), but the section boundary has no effect of its own.

4.3 Expand and collapse demonstration

Figure 9 shows all four sections collapsed, each displaying its title and rule range. Figure 10 shows a mixed state in which Management and Protection and Cleanup are expanded while DMZ Services and Outbound Access are collapsed. Together with Figure 8 (all expanded), these show the expand and collapse behaviour. Collapsing a section hides its rules from view without deleting or reordering them, which is useful when working on one part of a large rulebase.

Figure 10 also displays the message "Missing cleanup rule - Unmatched traffic will be dropped and not logged" beneath rule 8, although rule 8 is shown in the same figure as Any/Any/Any, Drop, Log. The screenshot does not explain why this message appears and I did not investigate the cause, so I assert no explanation here. The configuration of rule 8 as displayed is unchanged from Figure 7.

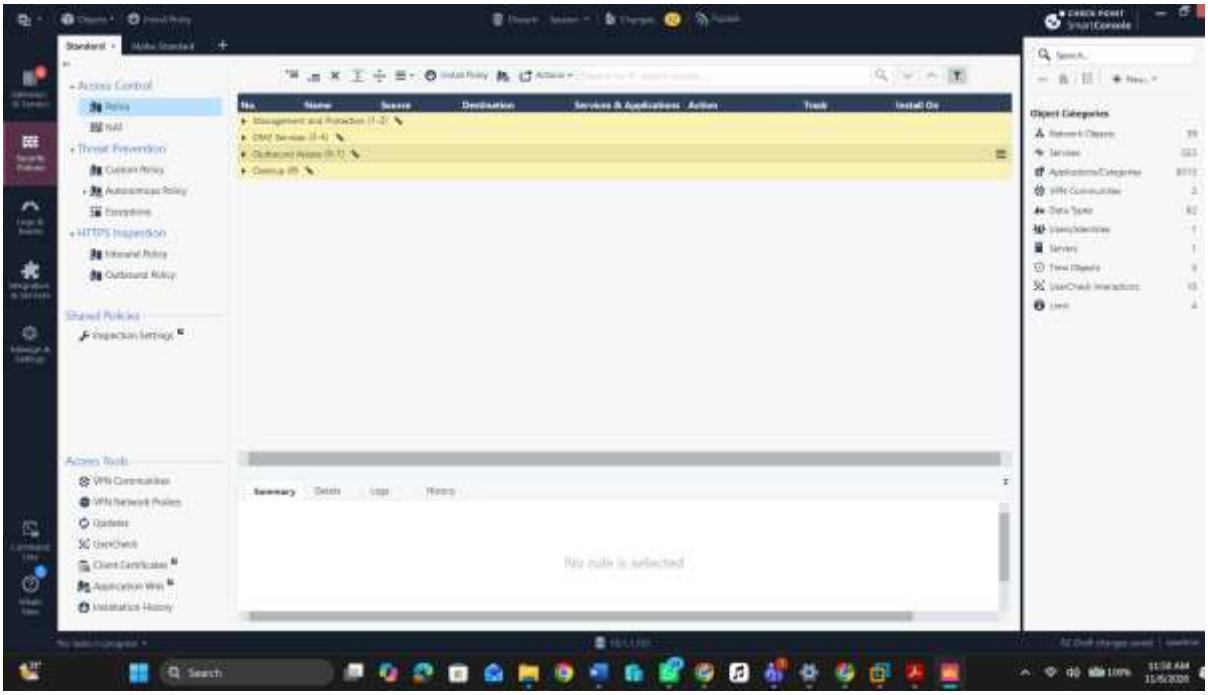


Figure 9: *All four sections collapsed, showing only the section titles and their rule ranges.*

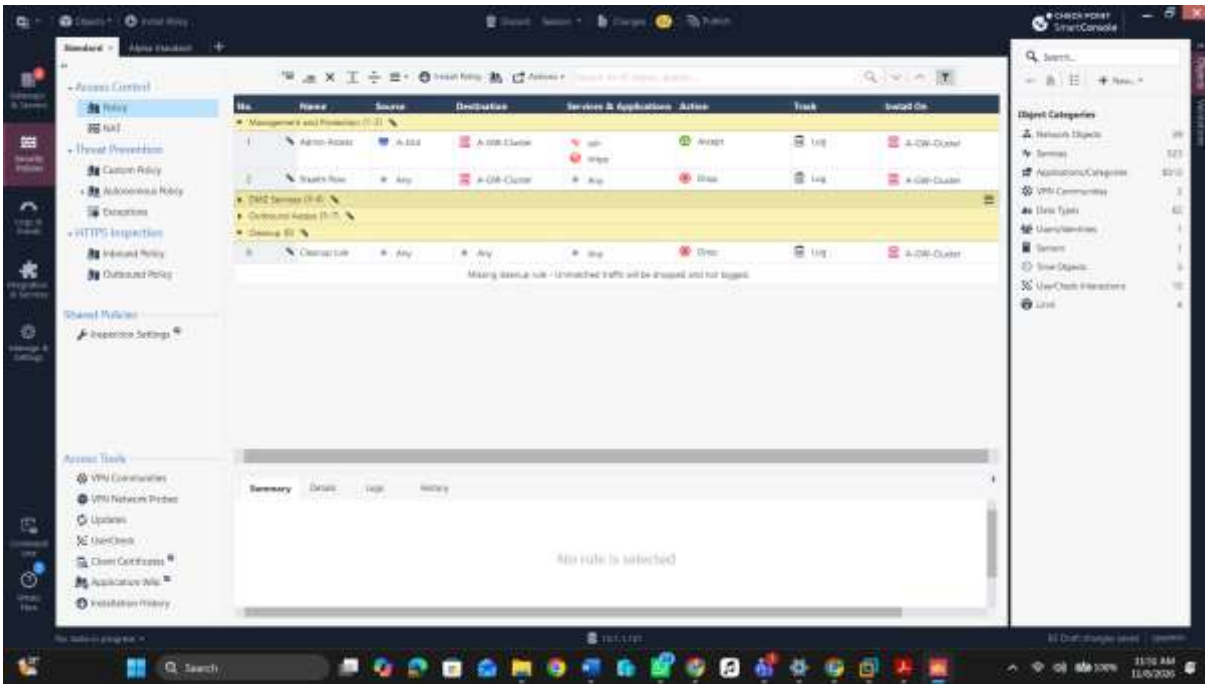


Figure 10: *Mixed view: Management and Protection and Cleanup expanded, DMZ Services and Outbound Access collapsed. The message below rule 8 is discussed in Section 4.3.*

5. Task 4: Configuring Hide NAT

5.1 The A-INT-NET NAT configuration

I opened the A-INT-NET network object (comment: Alpha-Site Internal Network) from the Object Explorer and configured its **NAT** page as shown in Figure 11. The values visible in the screenshot are listed in Table 7.

Table 7: NAT settings of the A-INT-NET object (Figure 11)

Setting	Value
Add automatic address translation rules	Enabled (ticked)
Translation method	Hide
Hide option selected	Hide behind the gateway
Hide behind IP address (alternative)	Not selected; IPv4 field shows 0.0.0.0
Install on gateway	All

"Hide behind the gateway" means that the translated source address is taken from the corresponding Security Gateway's interface, rather than from an address typed into the object (Check Point Software Technologies, n.d.c). The practical effect is that many private internal hosts appear to the outside world as the single address of the gateway.

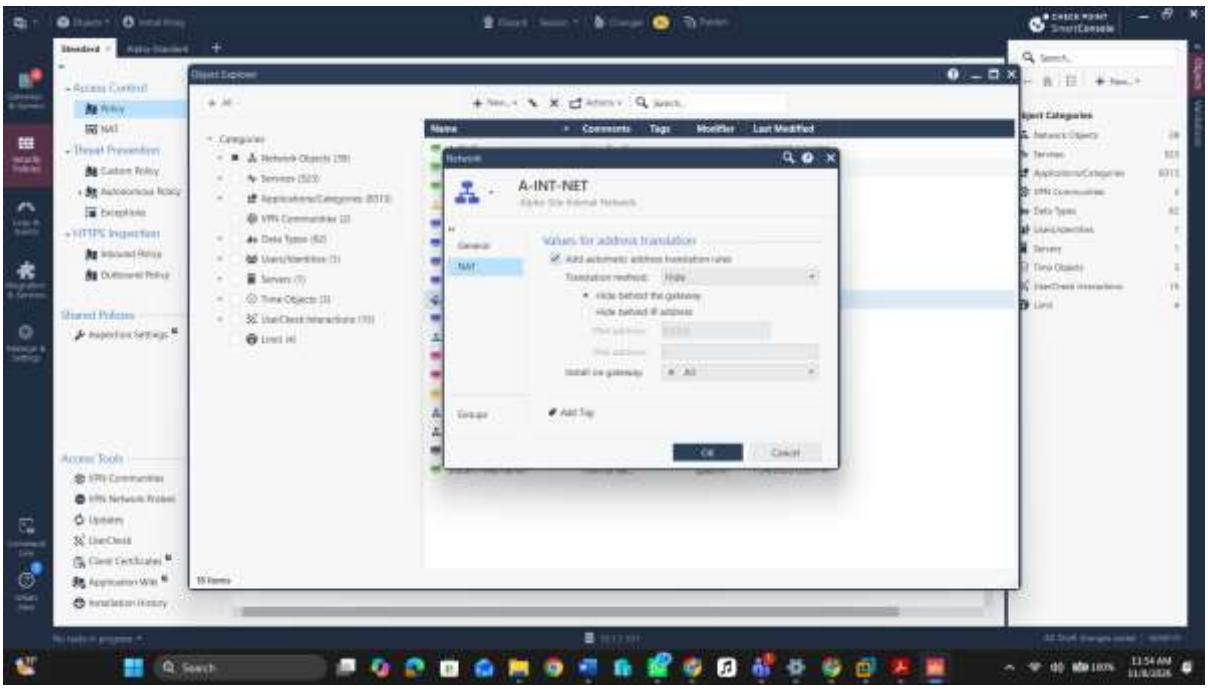


Figure 11: *Network object A-INT-NET, NAT page: automatic address translation rules enabled, method Hide, Hide behind the gateway, Install on gateway All.*

5.2 The automatically generated NAT rules

I then opened the NAT rulebase (Security Policies, Access Control, NAT). Figure 12 shows the section **Automatic Generated Rules: Network Hide NAT (1-6)**, containing six rules. Those that relate to A-INT-NET are rules 1 and 2:

Table 8: Automatic Hide NAT rules generated for A-INT-NET (Figure 12)

No.	Original Source	Original Destination	Original Services	Translated Source	Meaning
1	A-INT-NET	A-INT-NET	Any	Original	Traffic between hosts of the same network is not translated
2	A-INT-NET	Any	Any	A-INT-NET (Hiding Address)	Source of connections leaving A-INT-NET is hidden behind the gateway address

Rules 3 to 6 in the same section belong to other objects (A-MGMT-NET and the CP_default_Office_Mode_addresses_pool) and were not part of this task. The Translated Destination and Translated Services columns show Original for all rules, so only the source is translated.

The translated source is displayed in the screenshot as **A-INT-NET (Hiding Address)**. SmartConsole uses this notation for an object hidden behind its gateway; the actual address used is the one defined by the gateway interface, which the NAT rulebase does not display.

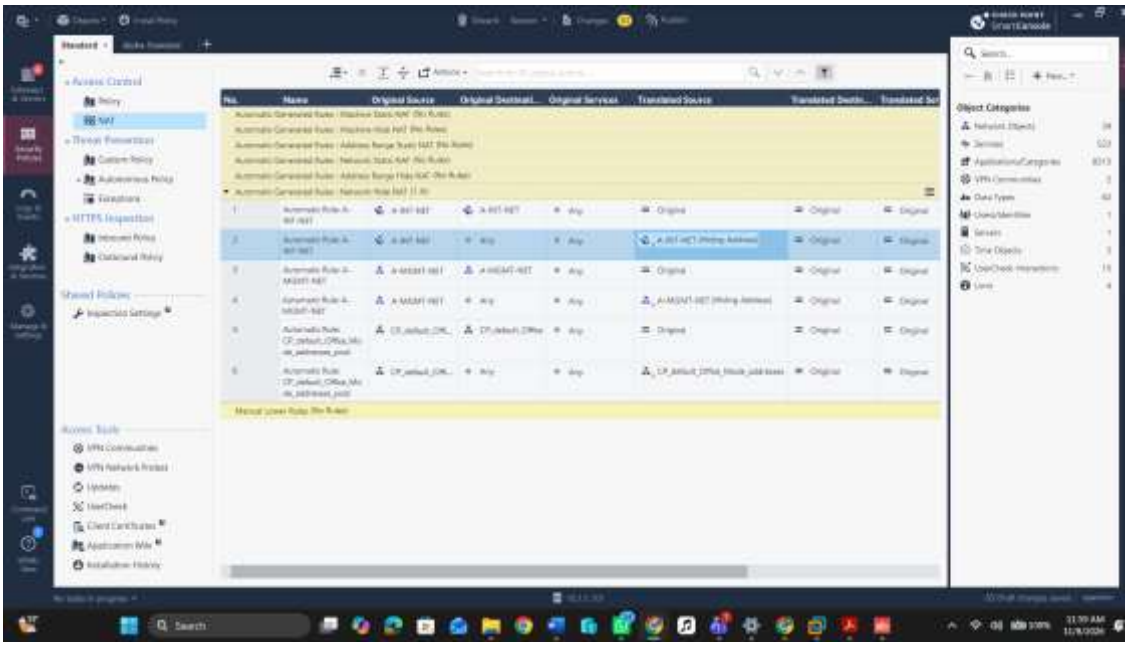


Figure 12: *NAT rulebase: Network Hide NAT (1-6). Rule 2 translates Original Source A-INT-NET to A-INT-NET (Hiding Address) for destination Any; rule 1 leaves A-INT-NET to A-INT-NET traffic untranslated. 83 draft changes are saved.*

5.3 Significance: automatic versus manual rules, many-to-one translation and ports

Why the rules are automatic. I did not write these rules in the NAT rulebase. The Security Management Server generated them from the NAT settings on the object itself, which is why they appear in a read-only section headed "Automatic Generated Rules" and carry names of the form "Automatic Rule: A-INT-NET". To change or remove them, I would change or disable NAT on the object, not in the rulebase. A manual NAT rule, by contrast, is created directly in the NAT rulebase and can match arbitrary combinations of source, destination and service. Check Point generates automatic rules for gateway, host, network and address-range objects, and for network objects it also generates a rule that leaves traffic within the same object untranslated, as seen in rule 1 (Check Point Software Technologies, n.d.c).

Many-to-one translation. Hide NAT translates many internal addresses (here, the whole 192.168.11.0/24 network) into a single outside address. Because one address now represents many hosts, the gateway cannot identify the correct internal host from the address alone.

What the gateway does with the source port. To keep simultaneous connections distinct, the gateway also replaces the original source port with a port of its own choosing and records the pairing in its connection table. When the reply arrives addressed to the hiding address and that port, the gateway looks up the entry and forwards the packet to the correct internal host and original port. This port translation is what allows many hosts to share one address.

Hide NAT versus Static NAT. Static NAT is one-to-one: each internal address maps permanently to one external address, in both directions, so an external client can initiate a connection to the translated address of an internal server. Check Point generates two rules for static NAT, translating the source of outbound traffic

and the destination of inbound traffic. Hide NAT is many-to-one and translates only the source of outbound connections; connections initiated from outside towards the hidden network are not supported (Check Point Software Technologies, n.d.c). Hide NAT therefore suits client networks such as A-INT-NET, while Static NAT suits servers, such as a DMZ web server that must be reachable from the Internet.

5.4 Configuration evidence versus live verification

Figures 11 and 12 are evidence that Hide NAT was **configured** on the management server and that the automatic rules were generated. They are not evidence that the gateway is translating traffic. NAT rules only take effect once the policy containing them has been installed on the gateway, and every installation attempt failed (Section 6). I did not test any connection from the internal network to the Internet, and I observed no NAT translation entry or log record. The brief's final step for this task (publication) is consistent with the later screenshots, which show no pending changes (Figure 14), but I did not capture the moment of publication itself.

6. Task 5: Rule Order Experiment and Installation History

6.1 The temporary rule

The experiment tests how the position of a rule changes which rule matches a connection. I created a temporary rule for it, with the configuration in Table 9 (the Rule Name, Source, Action, Track and Install On values are visible in Figures 13, 15 and 16; Destination and Services show Any).

Table 9: The temporary rule TEMP Block Internal

Field	Configuration
Name	TEMP Block Internal
Source	A-INT-NET
Destination	Any
Services & Applications	Any
Action	Drop
Track	Log
Install On	A-GW-Cluster
Created by	cpadmin (11/8/2026 1:01 PM, management server time)

Timestamps shown inside SmartConsole come from the management server and run about one hour ahead of the client workstation clock visible in the screenshot taskbar (for example, the rule's creation time reads 1:01 PM while the workstation clock reads 12:02 PM in Figure 13). This is consistent across the screenshots and does not affect the analysis.

6.2 Stages performed

Table 10: Stages of the experiment against the brief's steps

Step	Brief requirement	What was done and the outcome	Evidence
1	Add TEMP Block Internal above Internal Web Out; publish and install; repeat test T4; note the matching rule	Rule created immediately above Internal Web Out and published. Installation attempted: **failed**. Test **not performed**.	Figure 13 (position 6, Internal Web Out at 7); Figure 14 (failure, 13:07:50)

Step	Brief requirement	What was done and the outcome	Evidence
2	Move TEMP below Internal Web Out; publish and install; repeat T4; explain the difference	Rule moved to immediately below Internal Web Out and published. Installation attempted: **failed**. Test **not performed**.	Figure 15 (moved, 1 unpublished change); Figure 16 (published, status bar "Policy installation - Standard Failed"); Figure 17 (second failure, 13:11:26)
3	Delete the temporary rule; publish and install again	Rule deleted, original order restored and published. I have no evidence of a third installation and do not claim one.	Figure 18 (no TEMP rule; status bar "Published")
4	Open Access Tools, Installation History; identify the installation that contained the temporary rule; open its read-only view	I have no screenshot of the Installation History list. I do have the Install Policy Details window for the second failed attempt.	Figures 19 and 20

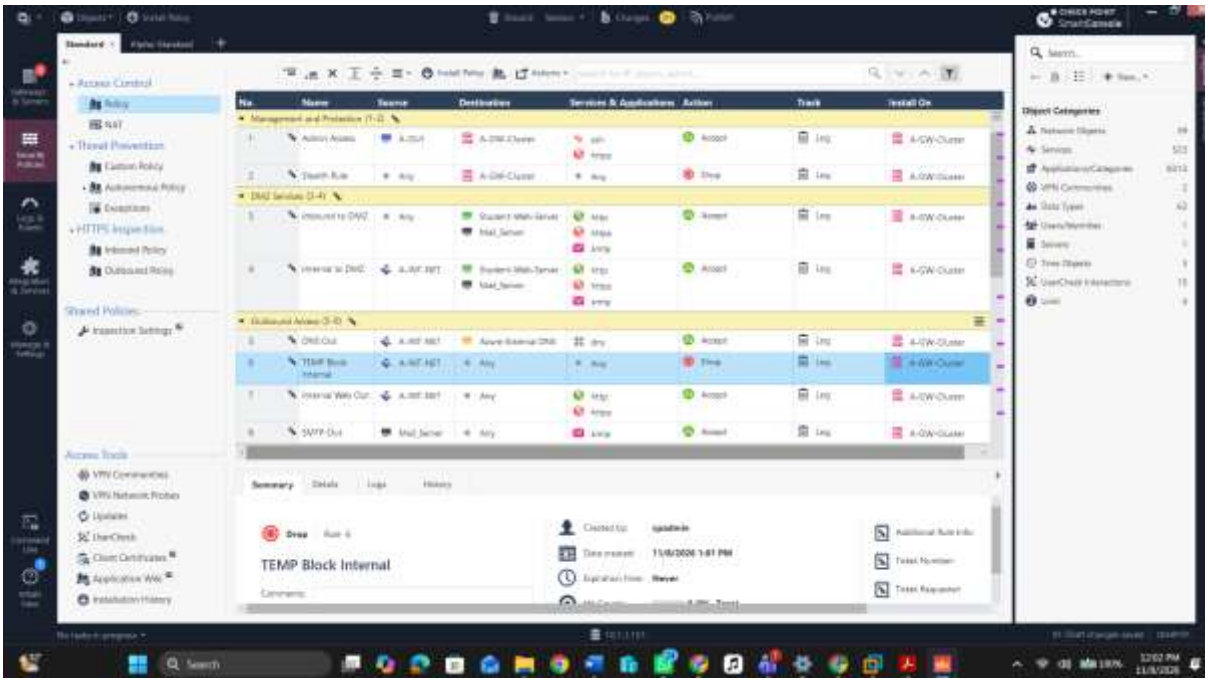


Figure 13: *TEMP Block Internal (rule 6: A-INT-NET, Any, Any, Drop, Log, Install On A-GW-Cluster) placed immediately above Internal Web Out (rule 7) in the Outbound Access section. The Summary pane shows a hit count of 0 (0%, Zero).*

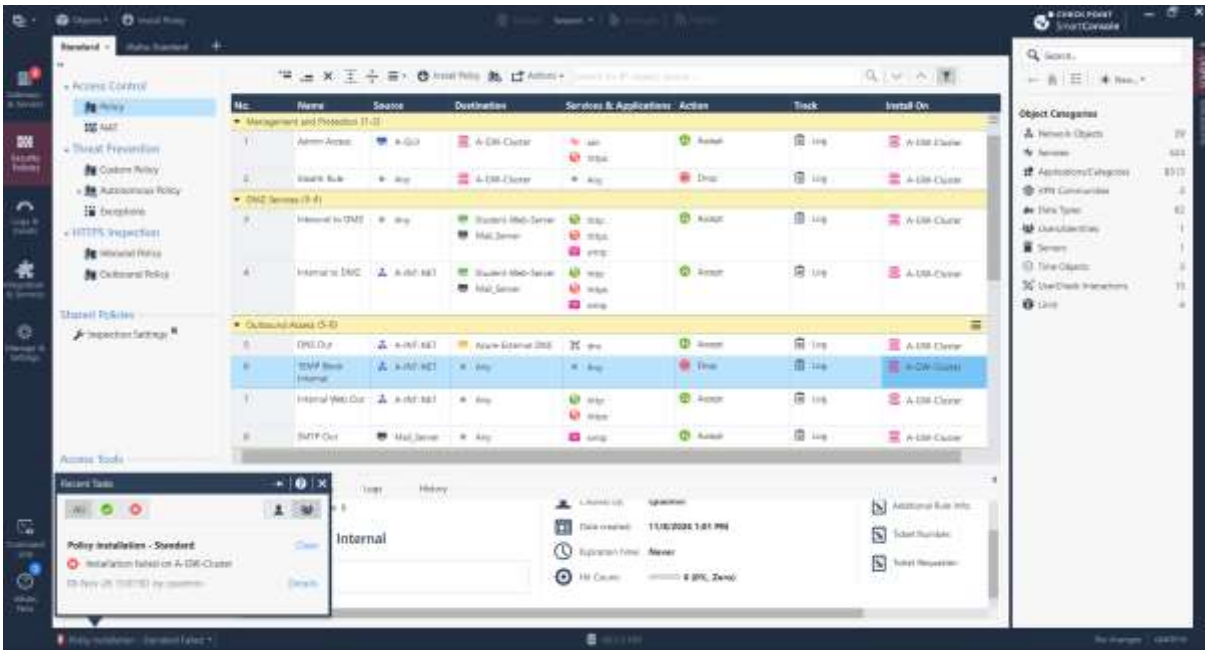


Figure 14: *Recent Tasks after the first attempt: "Policy installation - Standard", "Installation failed on A-GW-Cluster", 08-Nov-26 13:07:50 by cpadmin. The rulebase still shows TEMP Block Internal above Internal Web Out and no pending changes.*

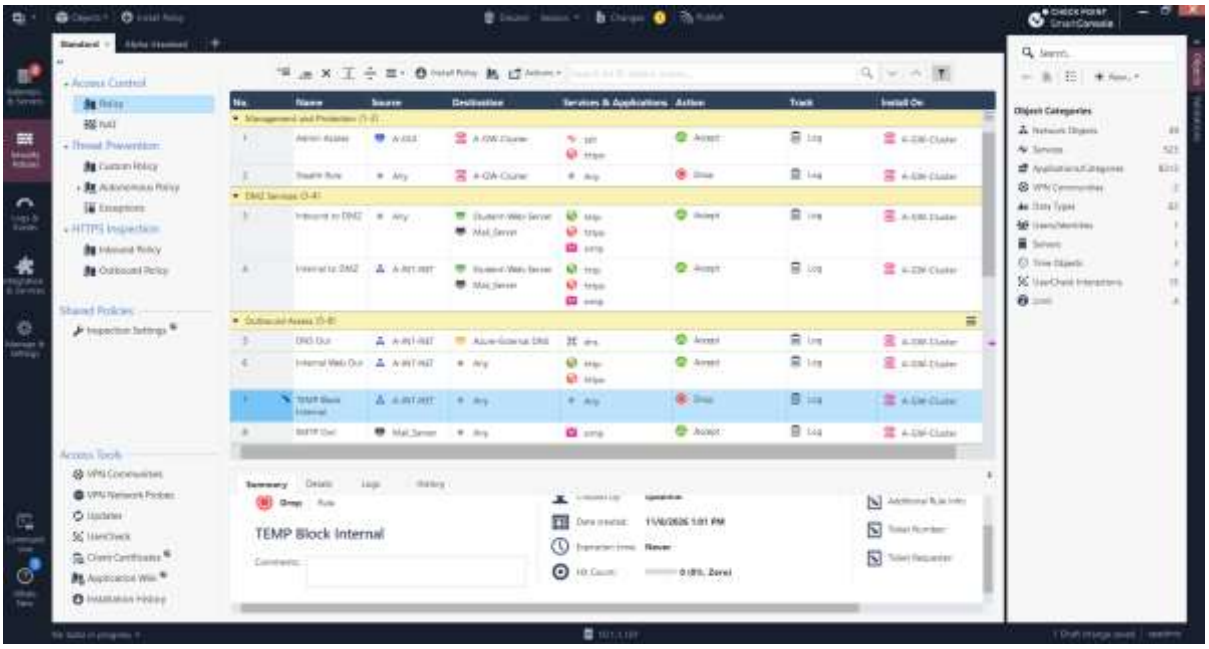


Figure 15: *TEMP Block Internal moved to rule 7, immediately below Internal Web Out (rule 6). The toolbar shows 1 draft change, so this view is before the second publication.*

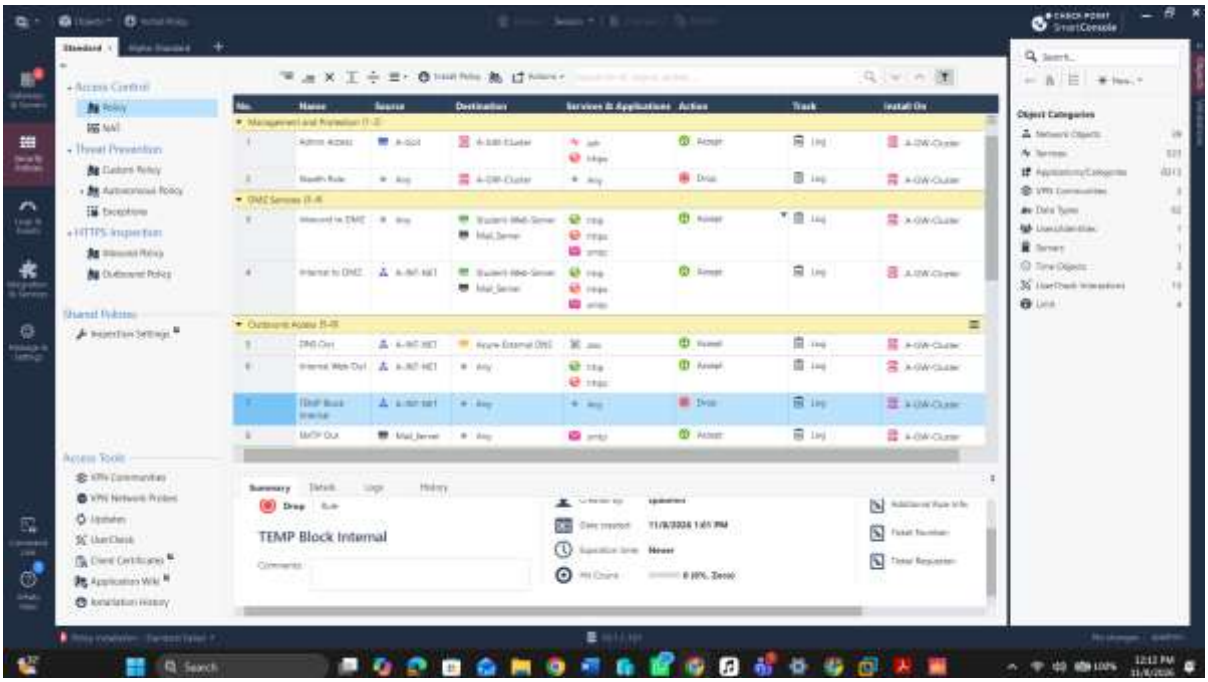


Figure 16: *After publication: no pending changes, TEMP Block Internal still at rule 7, and the status bar reads "Policy installation - Standard Failed".*

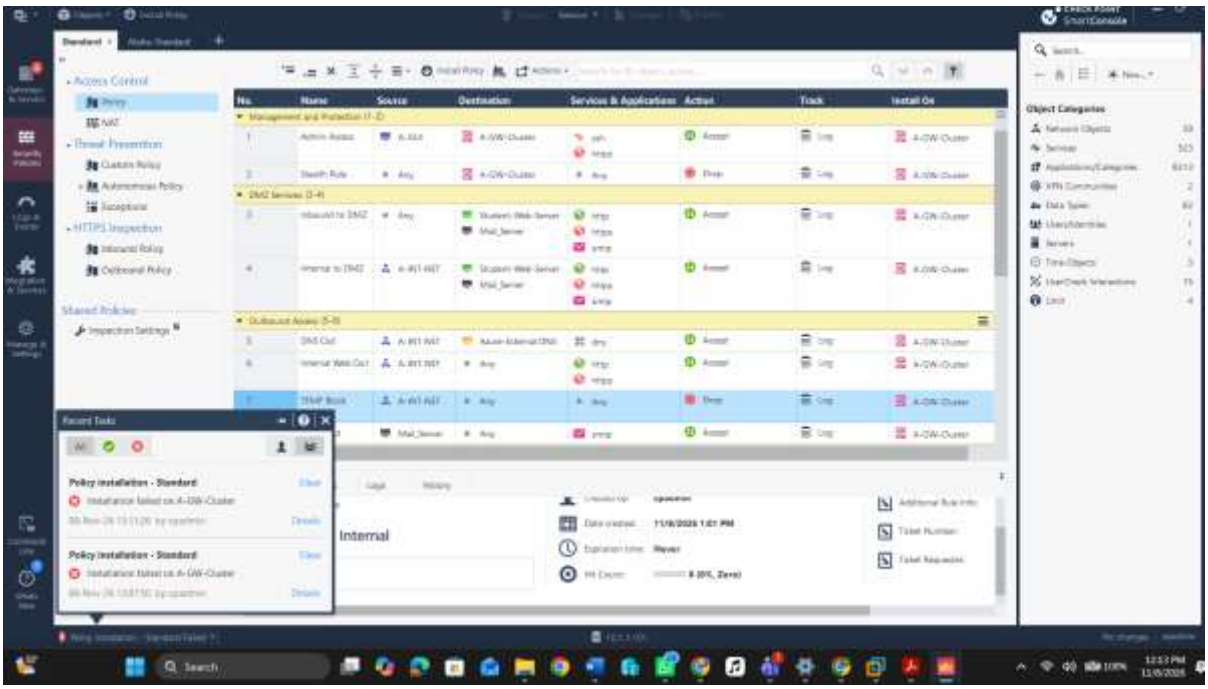


Figure 17: *Recent Tasks listing two failed installations of Standard on A-GW-Cluster, at 13:11:26 and 13:07:50, both by cpadmin. These are the two installations recorded for the experiment.*

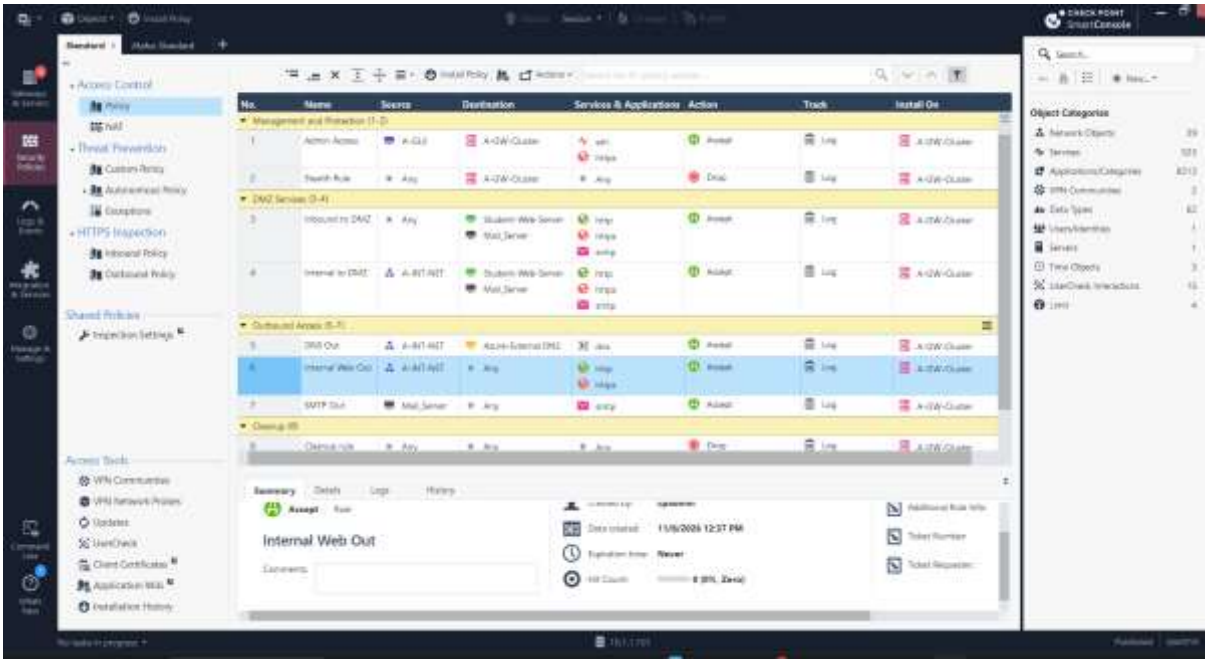


Figure 18: *The restored rulebase: TEMP Block Internal is gone and the original order is back (rules 1 to 8 under the four sections). The status bar reads "Published".*

6.3 Deletion and restoration of the original rulebase

After the second installation attempt I deleted the temporary rule and restored the original order. Figure 18 shows the rulebase with Outbound Access holding rules 5 to 7 (DNS Out, Internal Web Out, SMTP Out) and the Cleanup section holding rule 8, matching Figure 8. The Summary pane for Internal Web Out shows a creation date of 11/8/2026 12:37 PM (management server time), earlier than the temporary rule's 1:01 PM, which indicates that the original rule was retained rather than re-created. Its configuration (A-INT-NET, Any, http and https, Accept, Log, A-GW-Cluster) is unchanged.

6.4 The installation failure

Both installation attempts failed. Figure 19 shows the Install Policy Details window for the second attempt (13:11:26 to 13:11:36, ten seconds), initiated by cpadmin. The task status is "Installation failed on A-GW-Cluster" and the policy, Standard, is reported as Failed for all three gateway objects: A-GW-Cluster (10.1.1.1, version R82), A-GW-01 (10.1.1.2) and A-GW-02 (10.1.1.3). Expanding the A-GW-Cluster row (Figure 20) reveals the diagnostic messages in Table 11.

Table 11: Installation failure details (Figure 20)

Item	Value shown by SmartConsole
Task	Policy installation - Standard
Policy type / version	Access Control Policy / R82

Item	Value shown by SmartConsole
Message for A-GW-01	"The following certificate of gateway "A-GW-01" has already expired"
Message for A-GW-02	"The following certificate of gateway "A-GW-02" has already expired"
Certificate DN (both)	CN=A-GW-Cluster VPN Certificate,O=A-SMS.alpha.cp.g8f49h
Expiration date (both)	Sat May 23 07:13:37 2026
Reference given	sk31539

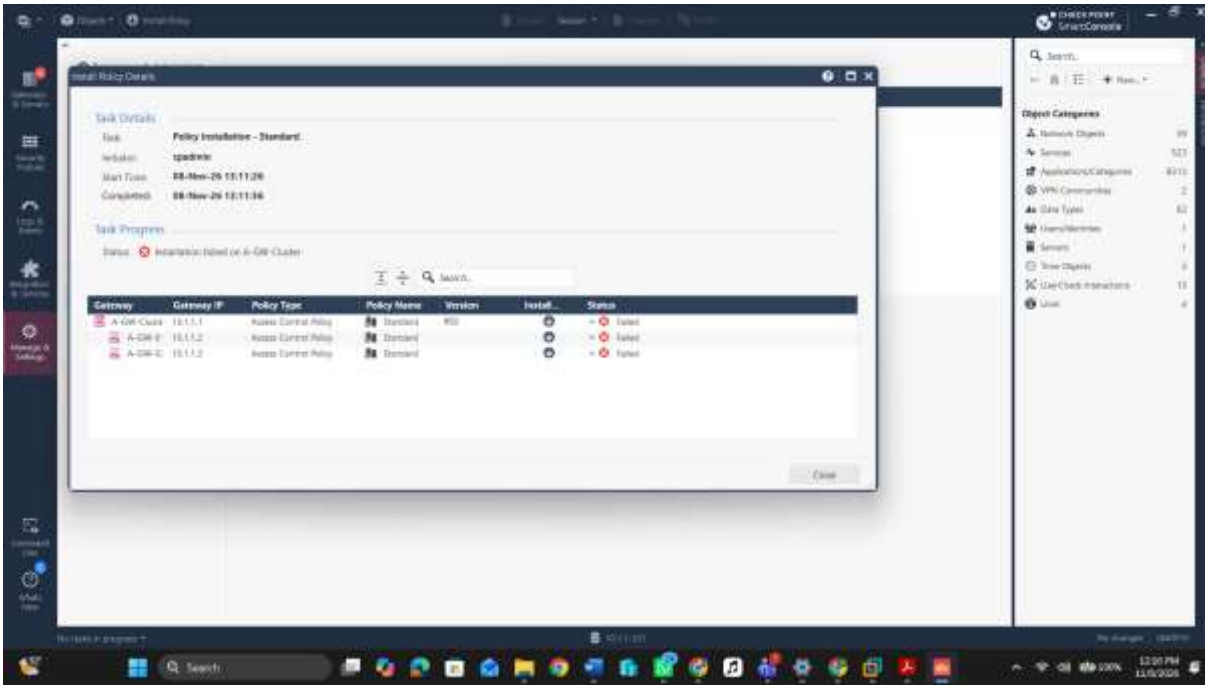


Figure 19: *Install Policy Details for the second attempt: Policy installation - Standard, 08-Nov-26 13:11:26 to 13:11:36, "Installation failed on A-GW-Cluster"; A-GW-Cluster, A-GW-01 and A-GW-02 all report Failed.*

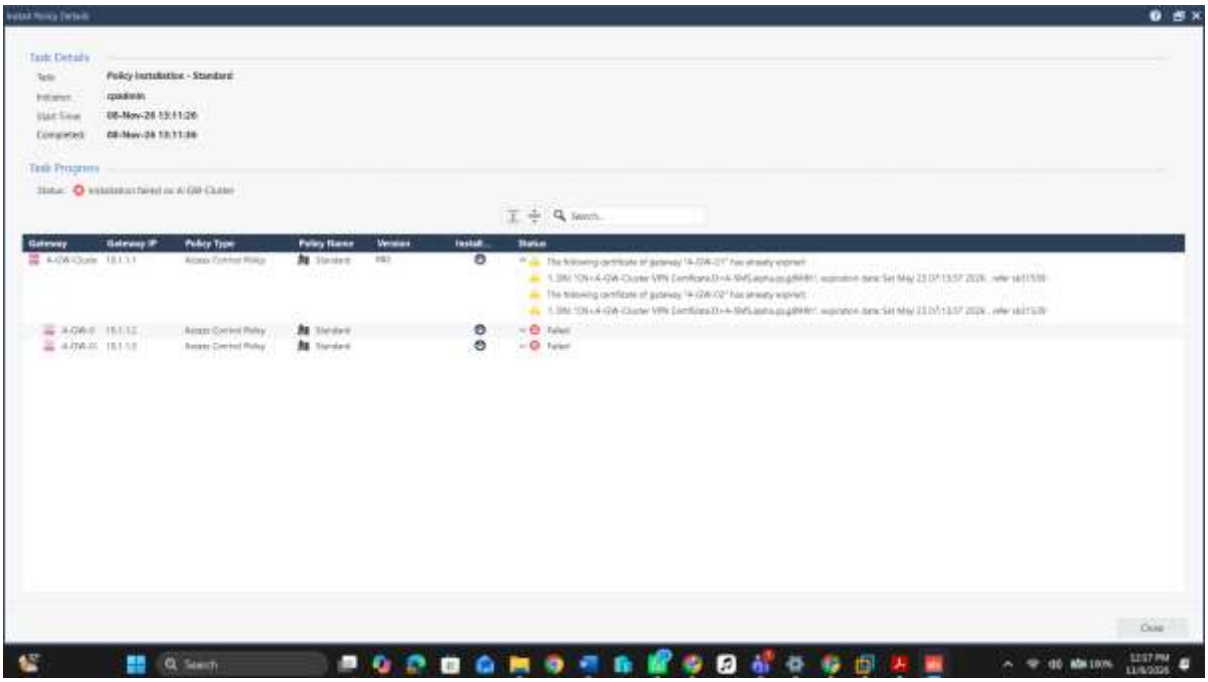


Figure 20: *The same window with the A-GW-Cluster row expanded: the VPN certificates of A-GW-01 and A-GW-02 (CN=A-GW-Cluster VPN Certificate,O=A-SMS.alpha.cp.g8f49h) "has already expired", expiration date Sat May 23 07:13:37 2026, with the reference sk31539.*

6.5 Implications of the expired certificates

The message identifies a gateway VPN certificate, whose distinguished name refers to the A-SMS management domain (O=A-SMS.alpha.cp.g8f49h), that had passed its expiry date. The date shown, 23 May 2026, is 169 days before the date of the installation attempts (8 November 2026). SmartConsole presents these two messages as the only diagnostics for the failed task, and SmartConsole refers to Check Point SK31539 for them. I could not retrieve SK31539 when preparing this report, so I do not summarise its content.

On the evidence available, the expired certificates prevented the policy from being installed on A-GW-Cluster. This is the interpretation of the messages SmartConsole displayed. I did not confirm it independently, for example from gateway-side logs, and I did not test it by renewing the certificates. The ten-second duration is consistent with the installation being rejected before the new policy was transferred to the members, although the screenshots do not state at which stage it stopped. Because the failure affected the cluster and both members, it was not specific to the rules being installed: the same result would be expected for any version of the Standard policy, including the final restored rulebase.

Renewal of the gateway VPN certificates and remediation of the gateways were outside the scope of the configuration activities I performed in this practical, and I did not attempt them.

6.6 Three distinct stages: publish, install, verify

Stage	What it does	Status in this practical
Publish	Commits the SmartConsole session's changes to the shared database on the Security Management Server, making them visible to other administrators.	**Completed.** Figures 14, 16 and 18 show no pending changes or "Published".
Install policy	Compiles the published policy and delivers it to the Security Gateways, which begin enforcing it.	**Failed** twice on A-GW-Cluster (Figures 14, 17, 19, 20).
Verify	Tests that traffic is allowed or dropped as designed and checks the resulting log entries.	**Not performed.** It depends on a successful install.

6.7 The rule-order experiment in principle

I could not run the traffic tests, so the following describes the expected behaviour from the first-match principle; I observed none of it. The test the brief calls T4 is not defined in the supplied brief. I interpret it here as an HTTP or HTTPS request from an internal host to an Internet destination, which is the traffic Internal Web Out exists to permit.

Position of TEMP Block Internal	Expected matching rule for internal HTTP/HTTPS to the Internet	Reason
Above Internal Web Out (Figure 13)	TEMP Block Internal (Drop, logged)	The gateway reaches TEMP first; the connection matches (A-INT-NET to Any, Any service) and is dropped. Internal Web Out is never reached.
Below Internal Web Out (Figures 15 to 17)	Internal Web Out (Accept, logged)	The gateway reaches Internal Web Out first; the connection matches and is accepted. TEMP is reached only by internal traffic not matched earlier.

The hit count of 0 shown for TEMP Block Internal (Figures 13 and 15) is consistent with the policy never having been enforced, but a zero count on its own does not prove it.

Why specific rules belong above general rules. The gateway applies the first rule that matches, so a broad rule placed above a narrow one swallows the traffic the narrow rule was meant to handle and the narrow rule never takes effect. Here the broad TEMP rule, which matches all internal traffic, would block web access that Internal Web Out was written to allow, whereas below it the broad rule only catches what the specific rules did not. Placing narrow exceptions first and general catch-all rules last keeps the intended permissions working and keeps the Cleanup rule, the most general rule, at the bottom.

7. Challenges, Limitations and Proposed Configurations

7.1 Gaps and inconsistencies in the supplied brief

The assignment brief is the primary source of the requirements. In several places it is incomplete or internally inconsistent. Where that materially affected the practical work, I disclose the limitation in Table 12 together with the approach I took. I have not altered the original requirements to make my work appear complete.

Table 12: Brief gaps and the approach adopted

Ref.	Gap or inconsistency	Approach adopted and effect
a	**Rule 4 is not defined.** The Task 2 table says Rule 4 is "added in Task 5", but Task 5 in the supplied brief is the rule-order experiment and nothing describes the rule.	I assumed a configuration and labelled it as proposed (Section 3.5). As configured, it is redundant behind Rule 3.
b	**Task numbering.** The supplied brief contains five tasks: 1 (explore), 2 (build rulebase), 3 (sections), 4 (Hide NAT) and 5 (rule order). The Task 2 table's reference to a "Task 5" that adds Rule 4 does not fit this numbering.	I use the task numbers printed in the supplied brief so that my headings match the assessed requirements: Task 3 (sections), Task 4 (NAT) and Task 5 (rule order).
c	**Test T4 is undefined.** Task 5 tells me to "repeat test T4", but no test T4 is described.	I interpreted it as internal-to-Internet web traffic (Section 6.7). Immaterial in practice, as no test could be run.
d	**Required evidence cannot be produced.** The brief asks for log screenshots for Task 5 steps 1 and 2, and for an Installation History screenshot.	I could not log any traffic because no policy was installed (Section 6). I supply Install Policy Details windows instead of an Installation History view.
e	**Security zones.** The brief's overview and Part B refer to security zones, but no practical task configures them.	I configured no zones; I address zones only in Part B, Question 10.
f	**Example addressing.** The brief's example addresses (10.1.1.0/24, 172.16.10.0/24 and others) do not match the lab topology.	As the brief instructs, I used the lab's own objects (Table 1).

7.2 Proposed configurations

- **Mail_Server host object (192.168.12.151).** No such object existed, so I created it as a proposed policy object (Section 3.1). I deployed and tested no mail server at this address, so Rules 3, 4 and 7, which reference it, describe intended policy only.
- **Rule 4, Internal to DMZ.** The configuration is an assumption (Section 3.5). It also illustrates a general point: a rule placed below a broader rule that already matches the same traffic has no effect.

7.3 Use of the Standard policy package

I carried out the work in the Standard package because it initially contained only the default Cleanup rule, which gave me a clean starting point for building the Access Control rulebase. The cluster's current Access Control policy, however, is Alpha-Standard (Figures 1 and 4). Had the installation succeeded, installing Standard on A-GW-Cluster would have replaced the policy the cluster currently enforces. That is acceptable in a disposable lab but would need change control in a production environment. This is a consideration only; no installation succeeded.

7.4 Expired certificates and the unsuccessful installation

Both installation attempts failed because SmartConsole reported expired gateway VPN certificates on A-GW-01 and A-GW-02 (Section 6.4). Certificate renewal was outside the scope of the activities I performed, so I did not change the cluster state. As a result, the rules in Table 5 and the Hide NAT rules in Table 8 exist only in the management database.

7.5 What could not be verified

- That A-GW-Cluster enforces the eight-rule policy, including the Stealth rule and the Cleanup rule.
- That internal hosts can browse, resolve DNS or reach the DMZ servers, or that external hosts can reach them.
- That Hide NAT translates the source addresses and ports of A-INT-NET traffic.
- The behaviour of the rule-order experiment, since neither traffic test was performed.
- That Rule 3 would pre-empt Rule 4 (Section 3.5). This follows from the first-match principle but was not observed.

8. Conclusion

What I configured. In the Standard policy package on A-SMS, I built an eight-rule Access Control rulebase following the Positive Control Model: a management rule, a Stealth rule, DMZ service rules, outbound rules, and a Cleanup rule that I changed from Accept to Drop with logging. I created a proposed Mail_Server host object to support the rules and added an assumed Rule 4 where the brief gave no definition. I organised the rules into four sections and configured Hide NAT on A-INT-NET, which produced automatic rules in the NAT rulebase.

What my evidence demonstrates. My screenshots demonstrate the default package and its Cleanup rule, the policy packages and Global Properties, the final rulebase and its sections (expanded and collapsed), the A-INT-NET NAT settings and generated NAT rules, the temporary rule at both positions, two failed installation tasks, and the restored rulebase. They also demonstrate the cause SmartConsole reported for the failures: expired gateway VPN certificates (23 May 2026) for A-GW-01 and A-GW-02.

What I could not verify, and why. Both of my installation attempts on A-GW-Cluster failed, so the policy was published on the management server but never installed on the gateway. Consequently, I could not verify gateway enforcement, the two rule-order traffic tests, log evidence for those tests, or live NAT translation. I have no evidence of a third installation after deleting the temporary rule, and my screenshots do not record the Installation History list, the hiding and showing of a column, or the editing of a section title. The practical therefore shows that I can design and configure a Check Point security policy and diagnose an installation failure, but not that I confirmed its operation on the gateway. I explain the rule-order principle from the first-match behaviour of the rulebase, not from observed results.

9. Part B: Theory Questions

Section B1: Multiple choice

Q1. Where is a Security Policy created, stored and enforced?

Answer: B. Created in SmartConsole, stored on the Security Management Server, enforced by Security Gateways. An administrator edits the policy in SmartConsole, which is only a client. The Security Management Server stores it in its database, and the Security Gateways enforce the policy installed on them. This is why, in the practical, publishing (to A-SMS) and installing (on A-GW-Cluster) are separate actions, and why the installation could fail while the published configuration remained intact.

Q2. Which control model does the Check Point Access Control firewall policy use?

Answer: A. Positive Control Model (whitelist). Only explicitly allowed traffic is permitted and everything not matched by an Accept rule is dropped, normally by a Cleanup rule at the bottom. A blacklist model would instead allow everything except what is listed. I changed the default Accept Cleanup rule in Figure 2 to Drop to implement the whitelist.

Q3. Where should the Stealth rule be placed?

Answer: C. As early as possible, typically immediately after the Management rules. It should be early so that no broader rule can accept traffic addressed to the gateway, but after the Management rule so that administrators are not dropped. It is rule 2, directly after Admin Access, in the practical. It is not placed at the bottom, because a general rule above it could accept the traffic first, and it is not above the Management rules, because it would block the administrator. Check Point's example policy places the Stealth rule directly after the administrator-access rule (Check Point Software Technologies, n.d.a).

Q4. Which statement about rulebase sections is correct?

Answer: D. Sections are visual divisions only and do not affect enforcement order. The gateway evaluates rules by position, whatever section they appear in (Section 4.2). They are not sent to the gateway as enforcement logic and have no relationship to zones.

Q5. What type of translation does Hide NAT perform?

Answer: B. Many-to-one. Many internal addresses share one translated address, and ports are translated to tell the connections apart (Section 5.3). Static NAT is the one-to-one type.

Section B2: Short answers

Q6. Explicit rules, Implied rules and the Implicit Cleanup Action

Explicit rules are the rules the administrator writes in the Access Control rulebase and that are visible there, such as the eight rules in Table 5. **Implied rules** are predefined by Check Point and are present in every policy so that the products can communicate, for example accepting control connections, SmartUpdate and Identity

Awareness control traffic (Figure 5). They are not shown in the normal rulebase; they can be viewed from the Actions menu of the Policy view as Implied Rules.

Where configured and placed. Implied rules are configured in **Global Properties, FireWall page**, where each can be enabled or disabled and given a position in the rulebase, such as First, Before Last or Last (Check Point Software Technologies, n.d.d; positions as shown in Figure 5). Changes require reinstalling the policy. Rules placed First are evaluated before the explicit rules, which is why gateway control traffic is not dropped by the Stealth rule. Logging of implied rules is off unless Log Implied Rules is ticked.

Implicit Cleanup Action. Every ordered layer has an implicit cleanup rule that handles traffic matching no explicit rule. Its default action is Drop, it can be set to Accept or Drop in the layer settings, and it is not displayed in the rulebase (Check Point Software Technologies, n.d.b). Because it is hidden, SmartConsole warns that traffic will be "dropped and not logged" without an explicit cleanup rule (Figure 10), so an explicit Cleanup rule with Track set to Log is added to record dropped traffic.

Q7. Why rule order matters

The gateway checks the rules from top to bottom and applies the **first** rule that matches; the rest are skipped for that connection. The same rules in a different order can therefore produce a different security outcome, so order must be designed rather than left as the sequence in which rules were created. Specific rules must precede general ones, and the catch-all rule must come last.

Example from the rulebase. Admin Access (rule 1) must stay above the Stealth Rule (rule 2). If the order were reversed, a connection from A-GUI to A-GW-Cluster over SSH would match the Stealth Rule first (Source Any, Destination A-GW-Cluster) and be dropped, and Admin Access would never be reached. A second example is the temporary rule of Task 5: TEMP Block Internal above Internal Web Out would drop internal web traffic, while below it would leave the web traffic accepted (Section 6.7), although this was not verified by testing.

Q8. Saving versus publishing, and installing

In SmartConsole, changes are made inside a **session**. SmartConsole saves the work as draft changes in that session (the status bar shows, for example, "82 Draft changes saved" in Figure 8), which protects against loss but is visible only to the administrator who owns the session. **Publishing** commits the session to the shared management database, making the changes visible to other administrators and releasing the objects the session had locked. Until then, the changes can still be discarded.

Installing applies a policy on the gateways. It works from the published database, so changes that remain only in an unpublished session are not part of the installed policy; in practice SmartConsole prompts the administrator to publish first. Publishing does not itself change gateway behaviour: that happens only at installation, which is the distinction demonstrated in the practical, where publication succeeded and installation failed.

Why organisations publish in a change window. After publication, the changes are in the shared database and can be installed by any administrator who next installs the policy, so a half-finished change could reach production unintentionally. Publishing only within an approved window ensures that the rulebase is complete, reviewed, and deployed when staff are available to monitor results and roll back, and it creates a clear audit point for the change.

Q9. Global Properties compared with Policy Packages

	Global Properties	Policy Packages
Scope	Settings that apply across the whole management domain and its gateways.	A named set of policies (Access Control, Threat Prevention and so on) installed together on chosen target gateways.
Example of what it controls	The implied rules and their positions (Figure 5); also NAT, VPN and authentication defaults.	Which rules a gateway enforces and which gateways are targets: Standard targets All gateways and Alpha-Standard targets A-GW-Cluster (Figure 4).
How to open	SmartConsole main menu (called the Applications menu in the brief), then **Global Properties**.	SmartConsole main menu, then **Manage policies and layers**, or the policy tabs in the Security Policies view.

In short, Global Properties set how the Check Point system behaves in general, whereas a policy package defines the particular security policy a set of gateways enforces.

Q10. Security Zones

The four predefined Security Zones are **WirelessZone**, **ExternalZone**, **DMZZone** and **InternalZone** (Check Point Software Technologies, n.d.e).

Two rules for assigning gateway interfaces. (1) A Security Gateway interface can belong to only one Security Zone. (2) Interfaces to different networks can be in the same zone, and interfaces must be assigned to zones before zones can be used in the rulebase (Check Point Software Technologies, n.d.e).

A benefit. Zone-based rules simplify the policy, since one rule can apply to many gateways, and networks can be added to an interface without changing the rulebase (Check Point Software Technologies, n.d.e). For example, one rule from InternalZone to ExternalZone would serve every internal network on every gateway, whereas network-object rules would need updating each time a network is added. I configured no zones in this practical.

Section B3: Scenario questions

Q11. An Accept Any rule inserted above the Stealth rule

Two security problems. First, the new rule matches all traffic, so rules below it, including the Stealth rule, the specific rules and the Cleanup rule, can never be reached. The policy becomes allow-everything, the opposite of the Positive Control Model, and the Stealth rule no longer protects the gateway. Second, the gateway itself, and the networks behind it, are exposed: any host can attempt connections to the firewall's management services (SSH and HTTPS) and to every internal and DMZ system, and the specific rules no longer restrict who can reach what. A further effect is on visibility. Every connection is logged against the one broad rule, so the logs no longer show which specific rule permitted what, and the Cleanup rule logs nothing because it is never matched.

Correction. Delete the rule or, if it is to be kept for a reason, narrow it so that it is not Any/Any/Any and move it below the Stealth rule. Then publish and install the corrected policy, and review the logs and hit counts for the period when the rule was active, to find any traffic that should not have been accepted. A change-control review should confirm why the rule was added.

Rule allowed above the Stealth rule. A **Management (administrator-access) rule**, such as Admin Access in the practical, which allows only specified administrator hosts to reach the gateway over specific services (SSH, HTTPS). It must sit above the Stealth rule because the Stealth rule drops all connections to the gateway and would otherwise block the administrators who need to manage it. It is safe in that position only because it is narrowly scoped.

Q12. Staff cannot reach the Internet after Hide NAT is configured

(a) Two benefits of NAT. (1) It conserves public IPv4 addresses: many hosts on a private network such as 192.168.11.0/24 can share one public address. (2) It hides the internal addressing scheme from external networks, so internal hosts are not directly addressable from outside, and it allows a private range to be used regardless of the provider's addressing.

(b) Three checks.

- **Check that the policy containing the NAT rules was published and installed, and that the NAT rulebase shows the automatic rule for A-INT-NET.** Automatic NAT rules take effect only after installation on the gateway. In this practical every installation failed on A-GW-Cluster, so the NAT rules could not have been enforced, which is precisely the sort of failure that would leave staff without Internet access. Install Policy Details would show whether the installation succeeded.
- **Check the logs for the staff traffic and the rule that matched it.** In SmartConsole Logs & Events, filter on an internal source and look at the action and matching rule. A Drop by the Cleanup rule would show that no Access Control rule permits the traffic (for example, DNS Out allows only the Azure-External DNS server, so queries to another resolver would be dropped). A log entry with an Accept

and a translated source would show that the rules and NAT are working and the fault lies elsewhere. This matters because an access rule, rather than NAT, is often the cause.

- **Check routing, the NAT configuration and the gateway interfaces.** Confirm that the gateway has a route to the Internet, that the external interface and the hiding address are correct ("Hide behind the gateway" uses the gateway interface address), and that the NAT settings were applied to the right gateway (the object's Install on gateway value). Without a valid return route or reachable address, translated packets leave but replies never return. Name resolution should also be tested, since a user who cannot resolve names appears to have no Internet.

10. References

Check Point Software Technologies (n.d.a) *Creating a Basic Access Control Policy*. R82 Security Management Administration Guide. Available at: https://sc1.checkpoint.com/documents/R82/WebAdminGuides/EN/CP_R82_SecurityManagement_AdminGuide/Content/Topics-SECMG/Creating-a-Basic-Access-Control-Policy.htm (Accessed: 8 November 2026).

Check Point Software Technologies (n.d.b) *Ordered Layers and Inline Layers*. R82 Security Management Administration Guide. Available at: https://sc1.checkpoint.com/documents/R82/WebAdminGuides/EN/CP_R82_SecurityManagement_AdminGuide/Content/Topics-SECMG/Ordered-Layers-and-Inline-Layers.htm (Accessed: 8 November 2026).

Check Point Software Technologies (n.d.c) *Working with Automatic NAT Rules*. R81 Security Management Administration Guide. Available at: https://sc1.checkpoint.com/documents/R81/WebAdminGuides/EN/CP_R81_SecurityManagement_AdminGuide/Topics-SECMG/Working_with_Automatic_NAT_Rules.htm (Accessed: 8 November 2026)

Check Point Software Technologies (n.d.d) *Implied Rules*. R82 Security Management Administration Guide. Available at: https://sc1.checkpoint.com/documents/R82/WebAdminGuides/EN/CP_R82_SecurityManagement_AdminGuide/Content/Topics-SECMG/Implied_Rules.htm (Accessed: 8 November 2026).

Check Point Software Technologies (n.d.e) *Security Zones*. R82 Security Management Administration Guide. Available at: https://sc1.checkpoint.com/documents/R82/WebAdminGuides/EN/CP_R82_SecurityManagement_AdminGuide/Content/Topics-SECMG/Security-Zones.htm (Accessed: 8 November 2026).